$$T_f = \frac{30400 \text{ J mol}^{-1}}{28.4 \text{ J K}^{-1} \text{ mol}^{-1}}$$

$$T_f = 1070.4 \text{ K}$$

58.

SOLUTION :

Given

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

$\Delta H_C^0 = -2220.2 \text{ kJ mol}^{-1}$ -----(1)

$C + O_2 \rightarrow CO_2$

$\Delta H_f^0 = -393.5 \text{ kJ mol}^{-1}$ -----(2)

$H_2 + \frac{1}{2} O_2 \rightarrow H_2O$

$\Delta H_f^0 = -285.8 \text{ kJ mol}^{-1}$ ----- (3)

$3C + 4H_2 \rightarrow C_3H_8$

$\Delta H_C^0 = ?$

$(2) \times 3 \Rightarrow 3C + 3O_2 \rightarrow 3CO_2$

$\Delta H_f^0 = -1180.5 \text{ kJ}$ -----(4)

$(3) \times 4 \Rightarrow 4H_2 + 2O_2 \rightarrow 4H_2O$

$\Delta H_f^0 = -1143.2 \text{ kJ}$ -----(5)

$(4) + (5) - (1) \Rightarrow 3C + 3O_2 + 4H_2 + 2O_2 + 3CO_2 + 4H_2O \rightarrow 3CO_2 + 4H_2O + C_3H_8 + 5O_2$

$\Delta H_f^0 = -1180.5 - 1143.2 - (-2220.2) \text{ kJ}$

$3C + 4H_2 \rightarrow C_3H_8$

$\Delta H_f^0 = -103.5 \text{ kJ}$

Standard heat of formation of propane is $\Delta H_f^0 (C_3H_8) = -103.5 \text{ kJ}$

59.

S. No	Liquid	Boiling points (^{0}C)	ΔH (kJ mol^{-1})
1.	Ethanol	78.4	+ 42.4
2.	Toluene	110.6	+ 35.2

SOLUTION :

For ethanol :

Given :

$T_b = 78.4^0 \text{ C} = (78.4 + 273)$

$= 351.4 \text{ K}$

$\Delta H_V (\text{ethanol}) = + 42.4 \text{ kJ mol}^{-1}$

$$\Delta S_V = \frac{\Delta H_V}{T_b}$$

$$\Delta S_V = \frac{+ 42.4 \text{ kJ mol}^{-1}}{351.4 \text{ K}}$$

$$\Delta S_V = \frac{+ 42400 \text{ J mol}^{-1}}{351.4 \text{ K}}$$

$\Delta S_V = +120.66 \text{ J K}^{-1} \text{ mol}^{-1}$

For Toluene :

Given :

$T_b = 110.6^0 \text{ C} = (110.6+273)$

$= 383.6 \text{ K}$

$\Delta H_V \text{ (toluene)} = + 35.2 \text{ kJ mol}^{-1}$

$$\Delta S_V = \frac{\Delta H_V}{T_b}$$

$$\Delta S_V = \frac{+ 35.2 \text{ kJ mol}^{-1}}{383.6 \text{ K}}$$

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